

LFSR / PRBS

A linear feedback shift register is a shift chain with XOR taps feeding the next
MSB

Maximal starter

- Starter: four-bit maximal polynomial $x^4 + x + 1$, seed zero-zero-zero-one
- Step and watch feedback XOR the taps, the register shift, and PRBS bits accumulate
- Measure period, you should see fifteen before the seed returns
- Try the short polynomial $x^4 + x^2 + 1$ for contrast, period under fifteen
- Load all-zero and see the lock warning: feedback stays zero forever
- Never seed all-zero on a real LFSR

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / LFSR / PRBS stepper

INTERACTIVE TOOL

LFSR / PRBS stepper

Step a Fibonacci LFSR from a seed — see tap XOR, the PRBS bit, and why period tops out at 2^n-1 for a maximal polynomial.

Starter example: 4-bit maximal x^4+x+1 , seed 0001 — step and watch the PRBS stream; measure period vs 2^n-1 .

Load starter example

Challenges (1/22)

Quiz: LFSR. Linear feedback shift register acronym? Answer: LFSR

Answer

Show hint

Check

Next

Idle

quiz-lfsr

quiz-prbs

quiz-period

quiz-zero

starter

preset-max4

preset-short

preset-max5

step

seed

measure-max

measure-short

zero-lock

prbs-len

demo

explain

poly-short

code-fb

tap-highlight

seed-alt

max5-period

full-path

Core ideas

LFSR

Shift register + XOR taps = next state.

PRBS

Pseudo-random bit stream from a tap / LSB.

Period

Maximal poly → all nonzero states: 2^n-1 .

LFSR lab starter

learn_digital — lfsr lab

Workbook practice

- In the workbook track
- Compute period for n equals four with a maximal poly
- Sketch why all-zero is forbidden
- Explain one use case: PRBS for a serial link or BIST pattern
- Name one pitfall: non-maximal polynomial or bad seed choice

Pitfalls to watch

- Do not confuse PRBS with true random, it repeats after the period
- Maximal taps matter; a wrong polynomial shortens the cycle
- All-zero locks the register
- And remember: the browser lab is literacy
- Real designs still need tap polynomials from tables, seed logic

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one LFSR step with XOR feedback
- When you are ready, take the short quiz, then continue to the ripple-carry adder animator